

The Planck gap as a theorem: every marking protocol needs $F^2\tau^3 > 9m\kappa$

Let κ be the smallest product of resolution and recoil width over the marks a record model can make. Then for **every** finite protocol of marks, of any number, at any times, the falling parabola can be told from the inertial line over a duration τ in the worst case only if

$$F^2\tau^3 > 9m\kappa, \quad \text{equivalently} \quad \tau\Delta E = \frac{F^2\tau^3}{2m} > \frac{9}{2}\kappa, \quad A_{\text{inertial,fall}} = \frac{vF\tau^3}{6m} > \frac{3v\kappa}{F},$$

and a two-mark protocol decides as soon as $F^2\tau^3 > 36m\kappa$ when marks saturating κ are available (Theorems 1–2). The threshold duration is therefore pinned within a factor $4^{1/3}$ by κ alone, and the Planck gap of the recorded Galileo comparison is positive exactly when $\kappa > 0$ (Corollary 3). The same bound governs Democritus insertion: the marks inside a window of duration τ' , used by themselves, record the force only if $F^2\tau'^3 > 9m\kappa$ (Corollary 4). Any record model in which a probe's later position records fix its momentum to arbitrary accuracy has $\kappa = 0$ (Proposition 5); quantum kinematics has $\kappa \geq 2\hbar = h/\pi$ by the Busch–Lahti–Werner error–disturbance theorem (Proposition 6), so the quantum Galileo comparison needs $\tau\Delta E > 9\hbar$; Newton's optics gives the heuristic value $\kappa = \Lambda p = h/2$ of the mark-floor note. The proof is a separation argument for the adversary's feasibility set, an AM–GM step that turns $\delta\Delta \geq \kappa$ into a square root, and one integration by parts that converts the certificate into a Dirichlet energy. Constants are explicit throughout; nothing is fitted.

1. Setting and definitions

Transverse coordinate y , mass $m > 0$, comparison on $[0, \tau]$ between the inertial hypothesis I and the falling hypothesis F, which differ by the displacement

$$P(t) = \frac{Ft^2}{2m}, \quad P(0) = P'(0) = 0, \quad P'' = \frac{F}{m}.$$

A **mark** at time t_j has a **resolution** $\delta_j > 0$: the datum it leaves is an interval of length δ_j containing $y(t_j)$; and a **recoil width** $\Delta_j \geq 0$: the transverse impulse it delivers to the body lies, under either hypothesis, in an interval of length Δ_j that is fixed by the totality of records, so that two runs which agree in every record can differ in that impulse by at most Δ_j . Between marks the body moves freely under I and with the force F under F. A **protocol** is a finite set of marks at times $0 = t_0 < t_1 < \dots < t_k \leq \tau$; the mark at $t_0 = 0$ is the preparation, whose recoil width Δ_0 is the width within which the records fix the initial transverse momentum. A protocol **decides** the comparison in the worst case when no pair of runs, one under I and one under F, with initial positions, initial momenta and recoils allowed by the marks, produces the same data. Set

$$\kappa = \inf_j \delta_j \Delta_j$$

over the marks the model can make; a model satisfies the **mark trade-off** with constant κ when every available mark has $\delta\Delta \geq \kappa$. For the corpuscular model of the mark-floor note this is premise M3 with $\kappa = \Lambda p$.

Feasibility system. Write $y_F - y_I = P + d$. Then $d(t) = x + vt + \sum_{j \geq 1} \rho_j(t - t_j)_+/m$ with $x \in \mathbb{R}$ free (both initial positions lie in the same datum interval, so their difference is bounded only by the datum constraint at t_0), $|v| \leq \Delta_0/m$ and $|\rho_j| \leq \Delta_j$. Two runs give the same data exactly when

$$|d(t_i) + P(t_i)| \leq \delta_i \quad (i = 0, 1, \dots, k). \quad (1)$$

The protocol decides if and only if (1) has no solution $(x, v, \rho_1, \dots, \rho_{k-1})$ in the stated ranges. (The last mark's recoil affects nothing that is recorded and is omitted.)

2. Certificate of decidability

Lemma 1 (separation). The protocol decides if and only if there are multipliers $\mu_0, \dots, \mu_k \in \mathbb{R}$ with $\sum_i \mu_i = 0$ and

$$\sum_i \mu_i P(t_i) > \sum_i |\mu_i| \delta_i + \frac{\Delta_0}{m} |S_0| + \sum_{j \geq 1} \frac{\Delta_j}{m} |S_j|, \quad S_j = \sum_{i > j} \mu_i (t_i - t_j). \quad (2)$$

Proof. Let $L : (x, v, \rho) \mapsto (d(t_i))_i \in \mathbb{R}^{k+1}$ be the linear map of Section 1 and B the box $|v| \leq \Delta_0/m$, $|\rho_j| \leq \Delta_j$. The system (1) is feasible exactly when the closed convex set $C = \{L(x, v, \rho) : x \in \mathbb{R}, (v, \rho) \in B\}$ meets the compact box $Q = \{q : |q_i + P(t_i)| \leq \delta_i\}$. C is the sum of a line and a compact convex set, hence closed and convex. If $C \cap Q = \emptyset$, the separation theorem for a closed and a compact convex set gives μ with $\sup_{q \in Q} \mu \cdot q < \inf_{c \in C} \mu \cdot c$. Since x is free the infimum is $-\infty$ unless $\mu \cdot L(1, 0, 0) = \sum_i \mu_i = 0$; with that, $\inf_C \mu \cdot c = -\frac{\Delta_0}{m} |\sum_i \mu_i t_i| - \sum_{j \geq 1} \frac{\Delta_j}{m} |\sum_{i > j} \mu_i (t_i - t_j)|$ by choosing the signs of v and ρ_j , and $\sup_Q \mu \cdot q = -\sum_i \mu_i P(t_i) + \sum_i |\mu_i| \delta_i$. The strict inequality between them is (2); if the separation comes with the opposite orientation, replace μ by $-\mu$. Note $\sum_i \mu_i t_i = S_0$ when $\sum_i \mu_i = 0$. Conversely, if (2) holds and (x, v, ρ) solved (1), pairing (1) with μ would give $\sum_i \mu_i P(t_i) \leq \sum_i |\mu_i| \delta_i - \sum_i \mu_i d(t_i) \leq \sum_i |\mu_i| \delta_i + \frac{\Delta_0}{m} |S_0| + \sum_{j \geq 1} \frac{\Delta_j}{m} |S_j|$, contradicting (2). $\square$

Lemma 2 (trade-off to a square root). If $\delta_j \Delta_j \geq \kappa$ for all j , then the right side of (2) is at least $2\sqrt{\kappa/m} \sum_{j=0}^k \sqrt{|\mu_j| |S_j|}$.

Proof. For each j , $|\mu_j| \delta_j + \frac{\Delta_j}{m} |S_j| \geq 2\sqrt{|\mu_j| |S_j| \delta_j \Delta_j / m} \geq 2\sqrt{\kappa |\mu_j| |S_j| / m}$ by the arithmetic–geometric mean inequality; $S_k = 0$. $\square$

3. The universal lower bound

Lemma 3 (the certificate as a Dirichlet energy). Let $\sum_i \mu_i = 0$, $N(u) = \sum_{i: t_i > u} \mu_i$ and $T(u) = \int_u^\tau N(w) dw$ for $u \in \mathbb{R}$. Then T is continuous and piecewise linear, $T \equiv T(0)$ on $(-\infty, 0]$, $T \equiv 0$ on $[\tau, \infty)$, $T(t_j) = S_j$, the slope of T jumps by μ_j at t_j , and

$$\sum_i \mu_i P(t_i) = \frac{F}{m} \int_0^\tau T(u) du, \quad \sum_j \mu_j T(t_j) = - \int_0^\tau T'(u)^2 du. \quad (3)$$

Proof. N is piecewise constant, vanishes for $u < 0$ because $\sum_i \mu_i = 0$ and for $u \geq t_k$, and drops by μ_j as u crosses t_j ; so $T' = -N$ has the stated jumps and support. Also $T(t_j) = \sum_i \mu_i |\{u \in [t_j, \tau] : u < t_i\}| = \sum_{i > j} \mu_i (t_i - t_j) = S_j$. For the first identity, $P(t_i) = \int_0^{t_i} P'(u) du$ and Fubini give $\sum_i \mu_i P(t_i) = \int_0^\tau P'(u) N(u) du = - \int_0^\tau P' T' du = -[P' T]_0^\tau + \int_0^\tau P'' T du = \frac{F}{m} \int_0^\tau T du$, using $T(\tau) = 0$ and $P'(0) = 0$. For the second, T' is of bounded variation with jumps μ_j at t_j and no other variation, so $\sum_j \mu_j T(t_j) = \int_{\mathbb{R}} T dT' = [TT']_{-\infty}^\infty - \int T'^2 = - \int_0^\tau T'^2$, as T' vanishes outside $[0, \tau]$. $\square$

Theorem 1 (universal lower bound). Suppose every mark of a protocol satisfies $\delta_j \Delta_j \geq \kappa$. If the protocol decides the Galileo comparison on $[0, \tau]$ in the worst case, then

$$F^2 \tau^3 > 9 m \kappa.$$

Proof. By Lemma 1 there is μ with $\sum_i \mu_i = 0$ satisfying (2), and by Lemma 2 and (3)

$$\frac{F}{m} \int_0^\tau T du > 2\sqrt{\frac{\kappa}{m}} \sum_j \sqrt{|\mu_j| |S_j|} \geq 2\sqrt{\frac{\kappa}{m}} \sqrt{\sum_j |\mu_j| |S_j|} \geq 2\sqrt{\frac{\kappa}{m}} \sqrt{E}, \quad E = \int_0^\tau T'^2 du,$$

using $\sum_j \sqrt{a_j} \geq \sqrt{\sum_j a_j}$ for $a_j \geq 0$ and $\sum_j |\mu_j| |S_j| \geq |\sum_j \mu_j T(t_j)| = E$. The right side is nonnegative, so $\int_0^\tau T > 0$. Since $T(\tau) = 0$, $|T(u)| = |\int_u^\tau T'| \leq \sqrt{\tau-u} \sqrt{E}$ by Cauchy-Schwarz, hence $\int_0^\tau T \leq \int_0^\tau \sqrt{\tau-u} du \sqrt{E} = \frac{2}{3} \tau^{3/2} \sqrt{E}$, that is $\sqrt{E} \geq \frac{3}{2} \tau^{-3/2} \int_0^\tau T$. Inserting, $\frac{F}{m} \int_0^\tau T > 3\sqrt{\kappa/m} \tau^{-3/2} \int_0^\tau T$, and dividing by the positive integral, $F\tau^{3/2} > 3\sqrt{\kappa m}$. $\square$

The theorem uses nothing about the marks except the product bound: no least length, no least impulse separately, no restriction on how many marks are made or when, no assumption that the marks are of one kind, and no probabilistic model. In the mark-floor note's two-mark protocol the same inequality appeared with the constant 16; Theorem 1 says that no protocol whatever can push the constant below 9.

4. The matching upper bound and the gap

Theorem 2 (two marks suffice). Suppose the model offers, for the duration τ in question, a preparation mark with $\delta_0 = \sqrt{\kappa\tau/m}$ and $\Delta_0 = \kappa/\delta_0$, and a terminal mark with $\delta_1 \leq \sqrt{\kappa\tau/m}$. Then the two-mark protocol decides the comparison whenever $F^2\tau^3 > 36m\kappa$.

Proof. Under I the terminal position lies in an interval of length $w = \delta_0 + \Delta_0\tau/m = 2\sqrt{\kappa\tau/m}$; under F in its translate by $s(\tau) = F\tau^2/2m$. A datum of length δ_1 cannot meet both when $s(\tau) > w + \delta_1$, which holds if $F\tau^2/2m > 3\sqrt{\kappa\tau/m}$, that is $F^2\tau^3 > 36m\kappa$. $\square$

Corollary 3 (the Planck gap of the recorded comparison). Let $\tau_*(F, m)$ be the infimum of durations over which some available protocol decides the comparison. Under the hypotheses of Theorems 1–2,

$$9m\kappa \leq F^2\tau_*^3 \leq 36m\kappa, \quad \frac{9}{2}\kappa \leq \tau_*\Delta E(\tau_*) \leq 18\kappa, \quad \frac{3v\kappa}{F} \leq A(\tau_*) \leq \frac{12v\kappa}{F},$$

the lower bounds from Theorem 1 as infima and the upper bounds from Theorem 2.

In particular the recorded comparison has a positive floor if and only if $\kappa > 0$, and the floor is a fixed multiple of κ , between $\frac{9}{2}$ and 18 in the action-scaled variable $\tau\Delta E$. The geometric area inherits the factor v/F and has no floor of its own, which is the Q14 gate: the only action the record model contains is κ .

Corollary 4 (Democritus insertion). Let marks be inserted at $t_a = t_j < t_{j+1} < \dots < t_{j+r} = t_b$ inside $[0, \tau]$, and let the observer use these marks alone to decide whether the motion on $[t_a, t_b]$ was free or forced, with no information about the momentum at t_a beyond what these marks give. Then the window records the force only if $F^2(t_b - t_a)^3 > 9m\kappa$.

Proof. Shift the origin to t_a . With the momentum at t_a unknown, v is free as well as x , so the certificate of Lemma 1 acquires the extra condition $S_0 = \sum_i \mu_i t_i = 0$ and loses the term $\frac{\Delta_0}{m} |S_0|$, which was zero under that condition anyway. Lemmas 2–3 and the proof of Theorem 1 go through unchanged with τ replaced by $t_b - t_a$. $\square$

So inserted points finer than $\tau_* = (9m\kappa/F^2)^{1/3}$ show, by themselves, free motion only; the force reappears when enough of them are pooled that the pooled window exceeds τ_* , and it also reappears if momentum information from before the window is carried in, in which case the comparison is the whole-interval one and Theorem 1 applies to it. This is the rigorous form of the mark-floor note's Theorem 2, and of I003's remark 3: κ controls the mesh at which the marked polygon stops converging to Newton's curve.

5. What fixes κ

Proposition 5 (deterministic probes give $\kappa = 0$). Suppose the marks are made by probes that travel freely after the mark, that a probe's transverse momentum after the mark equals its momentum before minus the impulse delivered, and that a probe's position can be recorded at any distance D with a resolution δ_s bounded independently of D . Then for every mark $\Delta \leq p_{\parallel}(\delta + \delta_s)/D$ for all D , where $p_{\parallel}$ is the probe's longitudinal momentum, so $\Delta = 0$ and $\kappa = 0$.

Proof. The probe's transverse momentum after the mark is $p_{\parallel}$ times the tangent of its direction, which the two recorded positions fix to within $(\delta + \delta_s)/D$; momentum conservation transfers the same width to the delivered impulse; let $D \rightarrow \infty$. $\square$

This is the far-screen countermodel of the mark-floor note stated for any deterministic probe, and it is the mechanism behind the classical closures C068–C123 (synthesis): full-state records fix the recoil, the trade-off constant is zero, and Corollary 3 gives floor zero. A positive κ therefore requires that the probe's later position records **do not** fix its momentum, which is an indeterminacy of the probe's own kinematics.

Proposition 6 (quantum marks have $\kappa \geq 2\hbar$). Model a mark as an instrument on the body's Hilbert space $L^2(\mathbb{R})$ with a real output, and define its resolution and recoil width in the worst-case sense: for every input state whose position distribution is supported in an interval of length ℓ centred at x , the output is supported in an interval of length $\ell + \delta$ centred at x ; and for every input state whose momentum distribution is supported in an interval of length ℓ , the post-mark momentum distribution, shifted by the mark's nominal impulse, is supported in an interval of length $\ell + \Delta$. Then $\delta\Delta \geq 2\hbar$.

Proof. The calibration error of the output as an approximate position measurement, in the sense of Busch, Lahti and Werner (the supremum over sharply localized inputs of the Wasserstein-2 distance between output and input position distributions, PRL 111, 160405 (2013) and J. Math. Phys. 55, 042111 (2014), abstract and theorem statements), is at most $\delta/2$: a distribution supported within $\delta/2 + \ell/2$ of the input's support has Wasserstein distance at most that from it, and $\ell \rightarrow 0$. Likewise the calibration disturbance of momentum is at most $\Delta/2$, since a momentum measurement after the mark, corrected by the nominal impulse, is an approximate momentum measurement whose output lies within $\Delta/2 + \ell/2$ of the input momentum support. Busch–Lahti–Werner prove that the product of the position error and the momentum disturbance of any instrument is at least $\hbar/2$. Hence $(\delta/2)(\Delta/2) \geq \hbar/2$. $\square$

With Corollary 3 this gives the quantum Galileo comparison the rigorous floor

$$\tau_* \Delta E(\tau_*) \geq \frac{9}{2} \cdot 2\hbar = 9\hbar \approx 1.43 h,$$

while the N01 closed-loop benchmark, an interferometric protocol outside the class of marks considered here, needs $\frac{4}{3}\tau\Delta E \geq \pi\hbar$ for perfect discrimination, $\tau\Delta E \geq 3\hbar/8$. The two numbers bracket the same combination; the present one is a theorem for all mark protocols under the stated worst-case definitions, the other an exact optimum for one protocol and a probabilistic criterion.

The Newton-age value. The corpuscular premises M1–M3 of the mark-floor note give $\kappa = \Lambda p$, with $\Lambda = 1/89000$ inch measured by Newton and p unmeasured in his age; on the identification $p = h/\lambda$, $\Lambda = \lambda/2$ this is $\kappa = h/2$, within a factor $\pi/2$ of the proved quantum value h/π . Newton-age materials give Corollary 3 in full, once M3 is granted; Proposition 5 shows that Newton's own determinate fits deny M3 and give $\kappa = 0$; Proposition 6 shows that quantum kinematics grants it with an explicit constant. The logical structure of the Planck gap is therefore:

$$\text{Planck gap of the recorded Galileo comparison} > 0 \iff \kappa > 0 \iff \text{indeterminate probe kinematics},$$

with the first equivalence proved here for all finite mark protocols and explicit constants, and the second implication supplied by the quantum error–disturbance theorem. No step assumes a path-integral phase rule, a fixed measurement budget, or a supplied action unit other than κ itself; and κ is not supplied but characterized: it is zero for every deterministic probe with far-field position records and at least $2\hbar$ for every quantum instrument.

6. Consequence for STATE

The Planck gap is now a theorem with a converse: Theorem 1 and Theorem 2 sandwich the threshold of the recorded Galileo comparison between $9m\kappa$ and $36m\kappa$ in $F^2\tau^3$ for every finite protocol of marks, Corollary 4 does the same for Democritus insertion, Proposition 5 shows $\kappa = 0$ for every deterministic probe with far-field records, and Proposition 6 gives $\kappa \geq 2\hbar$ in quantum kinematics from an established measurement-uncertainty theorem. The premise that carries $\hbar > 0$ is thereby reduced from the corpuscular M3 to the single number κ , and the question “why is $\hbar > 0$ ” is the question “why is $\kappa > 0$ ”, answered negatively for deterministic probes and positively for quantum ones.

Open: (a) a consistency derivation of $\kappa > 0$ that does not start from quantum kinematics, for which Proposition 5 now states exactly what must fail, namely far-field position records fixing a probe’s momentum; (b) extension of Theorem 1 from the constant force to a general $P(t)$ with $P(0) = P'(0) = 0$, where the proof gives $\int_0^\tau P''T \leq \|P''\|_{L^2} \|T\|_{L^2}$ and hence $\|P''\|_{L^2(0,\tau)} \tau > \dots$; the constant-force case is the worst case for a given $\|P''\|_\infty$ and should be written out; (c) the probabilistic version, replacing worst-case widths by distributions, where the N01 phase threshold and C006 belong.